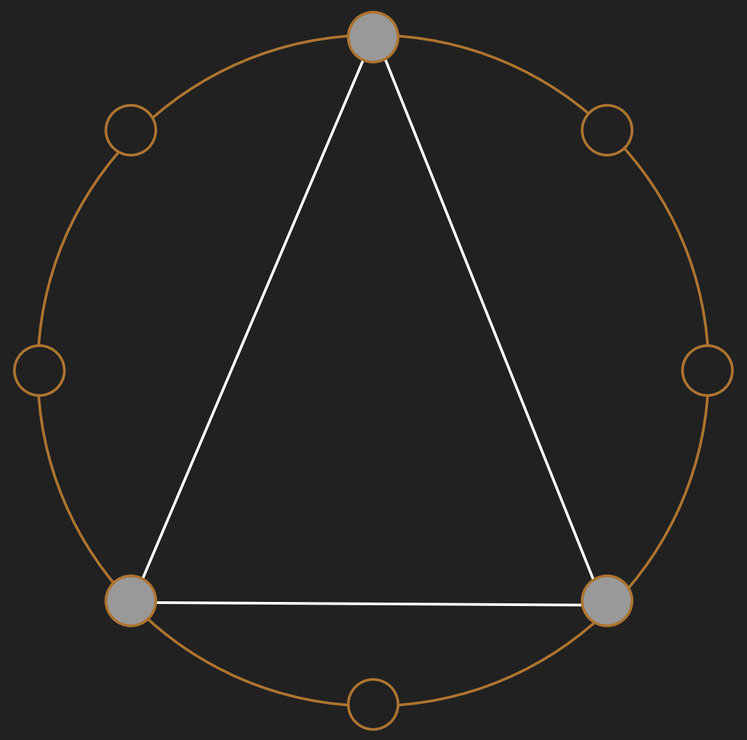


Euclidean Rhythms:

Prime numbers, cool rhythms, and odd time signatures

First described in 2005!

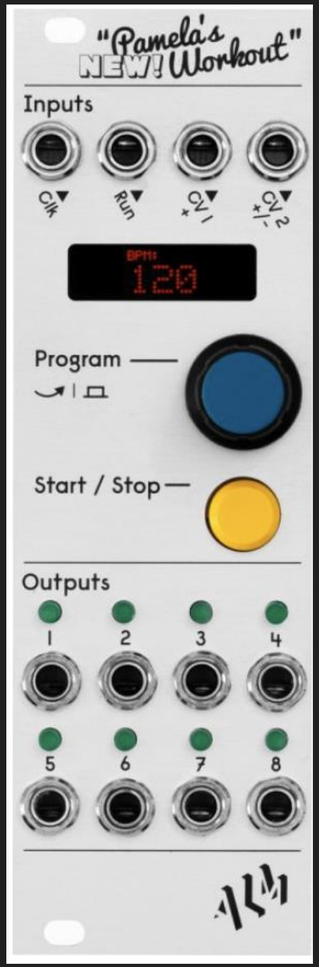
Literally

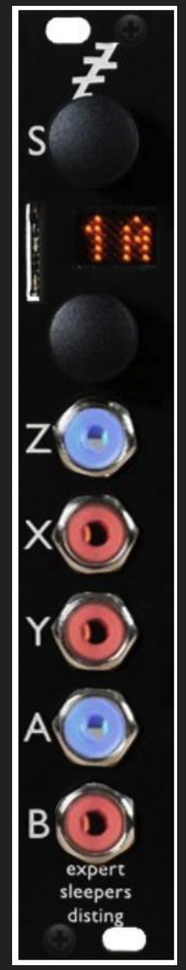
Euclidean
Circles v2
F
CLK
RST
vpme.de

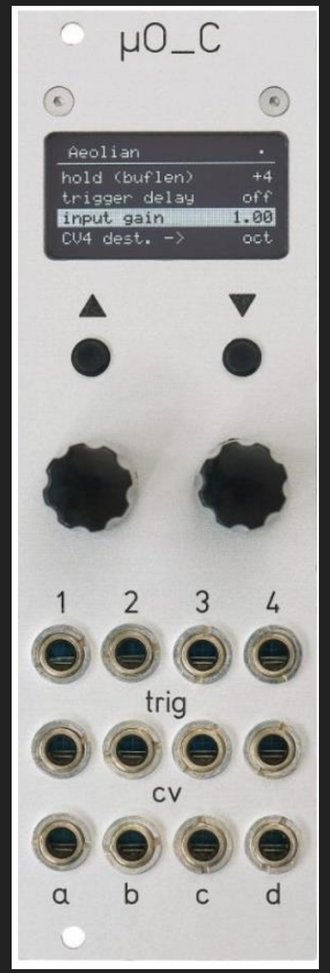
Euclid
TRIG
RESET
LENGTH
STEPS
LENGTH
STEPS
OFFSET
OUT
2hp

RUNNING ORDER
6hp
track 1 out
track 2 out
track rec. sel.
clock out**
live
reset 1
reset 2
REPEATS 2*
repeats 2 cv
REPEATS 1*
repeats 1 cv
clock in
RATE/DIV**
tuplets
track 1 mute
track 2 mute
hold both +* for patt.
long hold for euclid.
STEP
REST
ENDORPHINES

ΣΤΟΙΧΕΙΑ
α
β
α+β
REBEL TECHNOLOGY


"Pamela's NEW! Workout"
Inputs
Clk
Run
CV 1
CV 2
BPM
120
Program
Start / Stop
Outputs
1
2
3
4
5
6
7
8


S
1M
Z
X
Y
A
B
expert
sleepers
disting


μO_C
Aeolian
hold (buflen) +4
trigger delay off
input gain 1.00
CV4 dest. -> oct
1
2
3
4
trig
cv
a
b
c
d

TELETYPE
1
3
5
7
in
2
4
6
8
param
TR
1
2
3
4
CV
1
2
3
4
MONOME

But you can do all this with any old trigger sequencer

1:11 History & context
1:49 Divisors & relative primeness
4:50 Euclidean basics
9:02 Demos
12:18 Odd Time Signatures
14:57 Rhythm zoo

Slides at: https://bit.ly/euclideanrhythms

Euclid

"The Euclidean Algorithm"

c. 300 BCE

Eric Bjorklund

Timing of events in neutron accelerators

2003

Godfried Toussaint

"The Euclidean Algorithm Generates Traditional Musical Rhythms"

2005

The *Euclidean* Algorithm Generates Traditional Musical Rhythms

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Montréal, Québec, Canada
godfried@cs.mcgill.ca

Abstract

The *Euclidean* algorithm (which comes down to us from Euclid's *Elements*) computes the greatest common divisor of two given integers. It is shown here that the structure of the Euclidean algorithm may be used to generate, very efficiently, a large family of rhythms used as timelines (*ostinatos*), in sub-Saharan African music in particular, and world music in general. These rhythms, here dubbed *Euclidean* rhythms, have the property that their onset patterns are distributed as evenly as possible. *Euclidean* rhythms also find application in nuclear physics accelerators and in computer science, and are closely related to several families of words and sequences of interest in the study of the combinatorics of words, such as Euclidean strings, to which the *Euclidean* rhythms are compared.

1. Introduction

What do African rhythms, spallation neutron source (SNS) accelerators in nuclear physics, string theory (stringology) in computer science, and an ancient algorithm described by Euclid have in common? The short answer is: patterns distributed as evenly as possible. For the long answer please read on.

Mathematics and music have been intimately intertwined since the days of Pythagoras. However, most of this interaction has been in the domain of pitch and scales. For some historical snapshots of this interaction the reader is referred to H. S. M. Coxeter's delightful account [9]. Rhythm, on the other hand has been historically mostly ignored. Here we make some mathematical connections between musical rhythm and other areas of knowledge such as nuclear physics and computer science, as well as the work of another famous ancient Greek mathematician, Euclid of Alexandria.

2. Timing Systems in Neutron Accelerators

The following problem is considered by Bjorklund [5], [4] in connection with the operation of certain components (such as high voltage power supplies) of spallation neutron source (SNS) accelerators used in nuclear physics. Time is divided into intervals (in the case of SNS, 10 seconds). During some of these

http://cgm.cs.mcgill.ca/~godfried/publications/banff.pdf

Math Alert!

(But don’t worry about it)

If you’ve got this covered, you can skip forward to 4:34

$$\frac{4}{2} = 2$$

2 is a divisor of 4

$$\frac{5}{2} = 2\tfrac{1}{2}$$

2 is not a divisor of 5

$$\frac{0}{1} = 0 \qquad \frac{1}{1} = 1 \qquad \frac{2}{1} = 2 \qquad \frac{3}{1} = 3$$

1 is a divisor of everything

$$\frac{2}{2} = 1 \qquad \frac{3}{3} = 1 \qquad \frac{4}{4} = 1 \qquad \frac{5}{5} = 1$$

Every number is a divisor of itself

$$\frac{2}{2} = 1 \qquad \frac{3}{3} = 1 \qquad \frac{4}{4} = 1 \qquad \frac{5}{5} = 1$$

Every number is a divisor of itself

$$3 = 1 \times 3$$
$$4 = 1 \times 2 \times 2$$
$$5 = 1 \times 5$$
$$6 = 1 \times 2 \times 3$$

If a number only has 1 and itself as divisors, then it is prime

$$15 = 1 \times 3 \times 5$$
$$18 = 1 \times 2 \times 3 \times 3$$

3 is the greatest common divisor

$$10 = 1 \times 2 \times 5$$
$$21 = 1 \times 3 \times 7$$

1 is the greatest common divisor

10 & 21 are ‘relatively prime’

(That wasn’t so bad)

The only real take-away:

Some pairs of numbers have no common divisors - they are ‘relatively prime’

EUCLIDEAN BASICS

It’s all about distributing things as evenly as possible.

Onset = Event = On

3 variables describe a Euclidean rhythm

Number of steps
Number of events
Rotation/Offset

$$E(3, 8)$$

Number of events

$$E(3, 8)$$

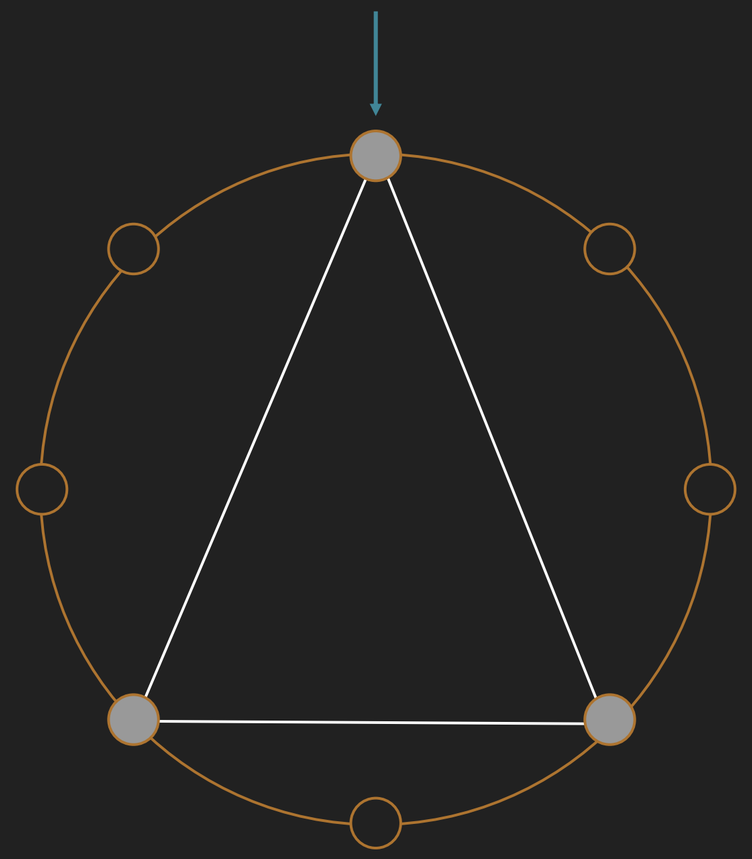
Number of steps

Number of events

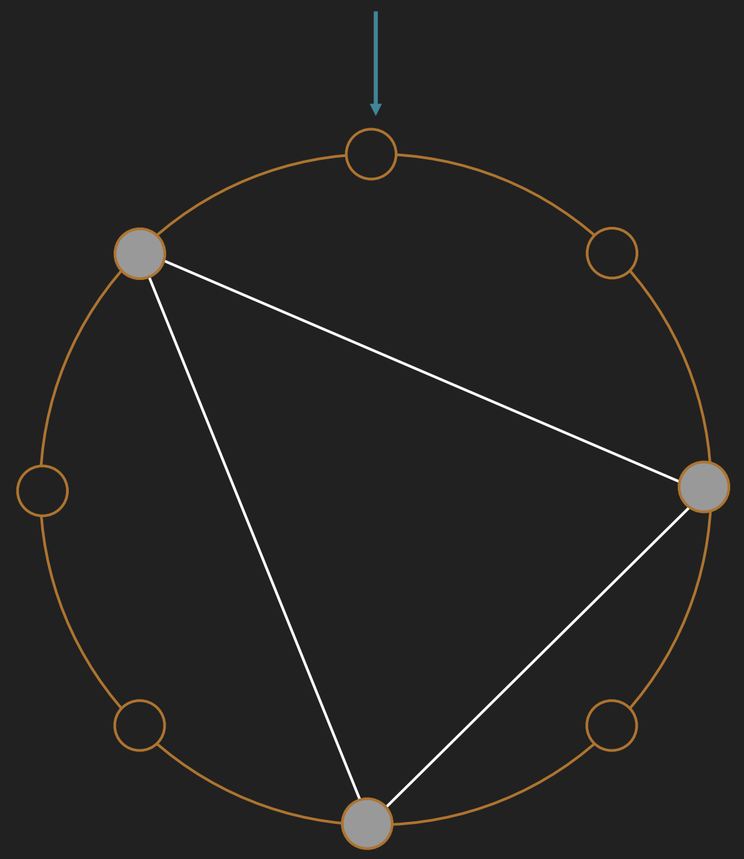
Number of steps

$$E(3, 8, 1)$$

Rotation



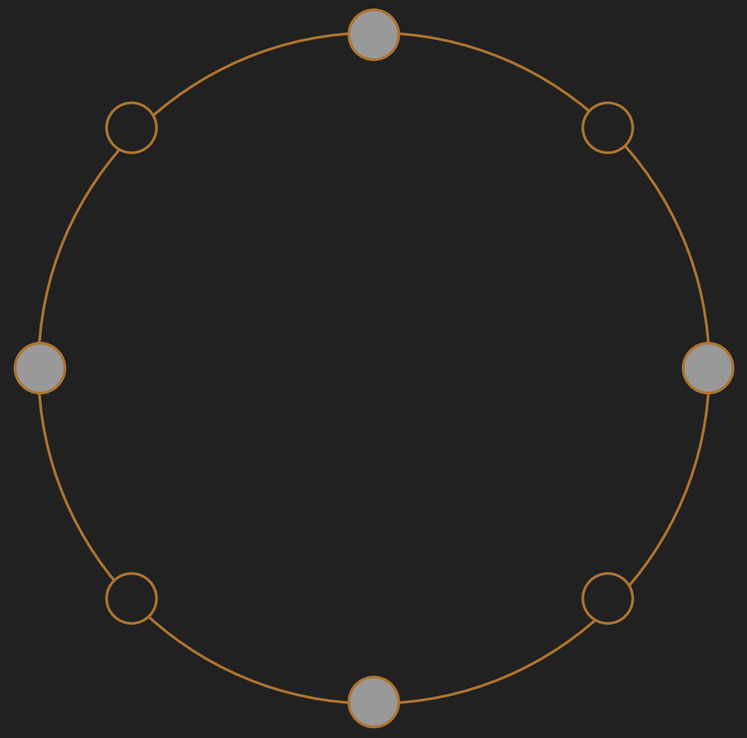
E(3,8,0)



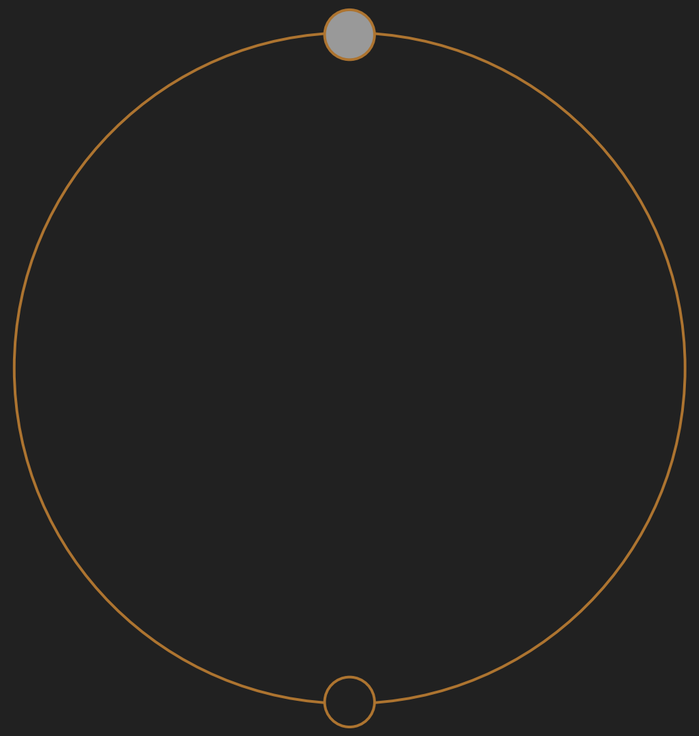
E(3,8,1)

3 variables describe a Euclidean rhythm

You can’t evenly divide 3 into 8

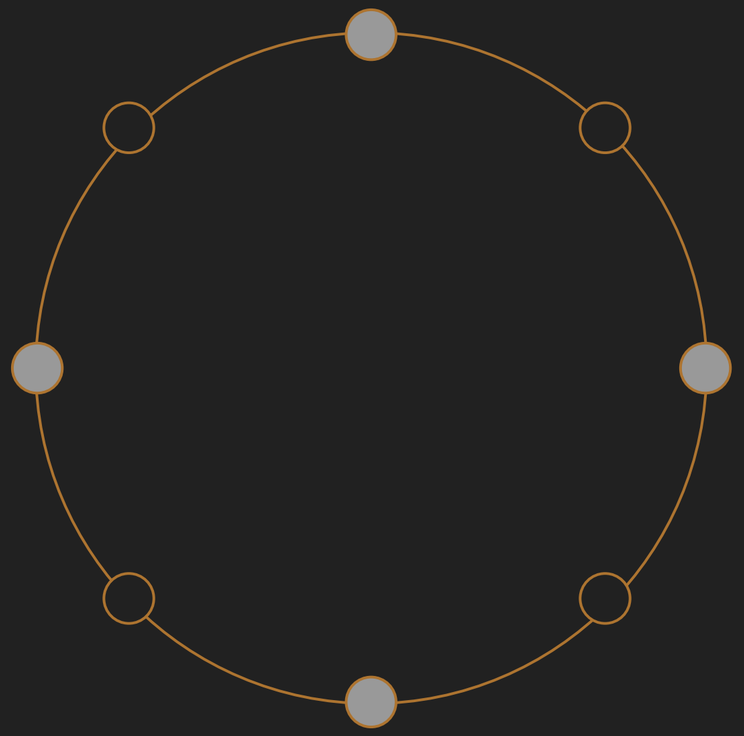


E(4,8,0)

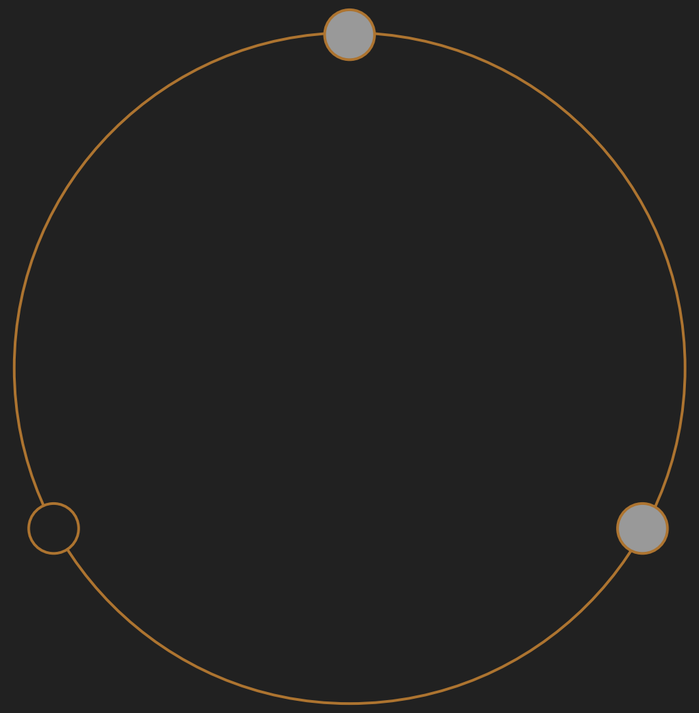


E(1,2,0)

x 4 =

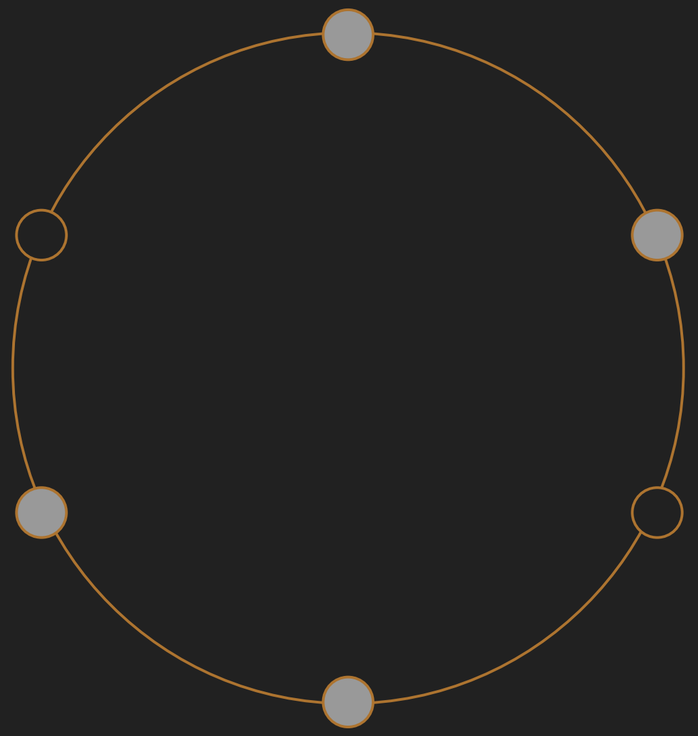


E(4,8,0)



E(2,3,0)

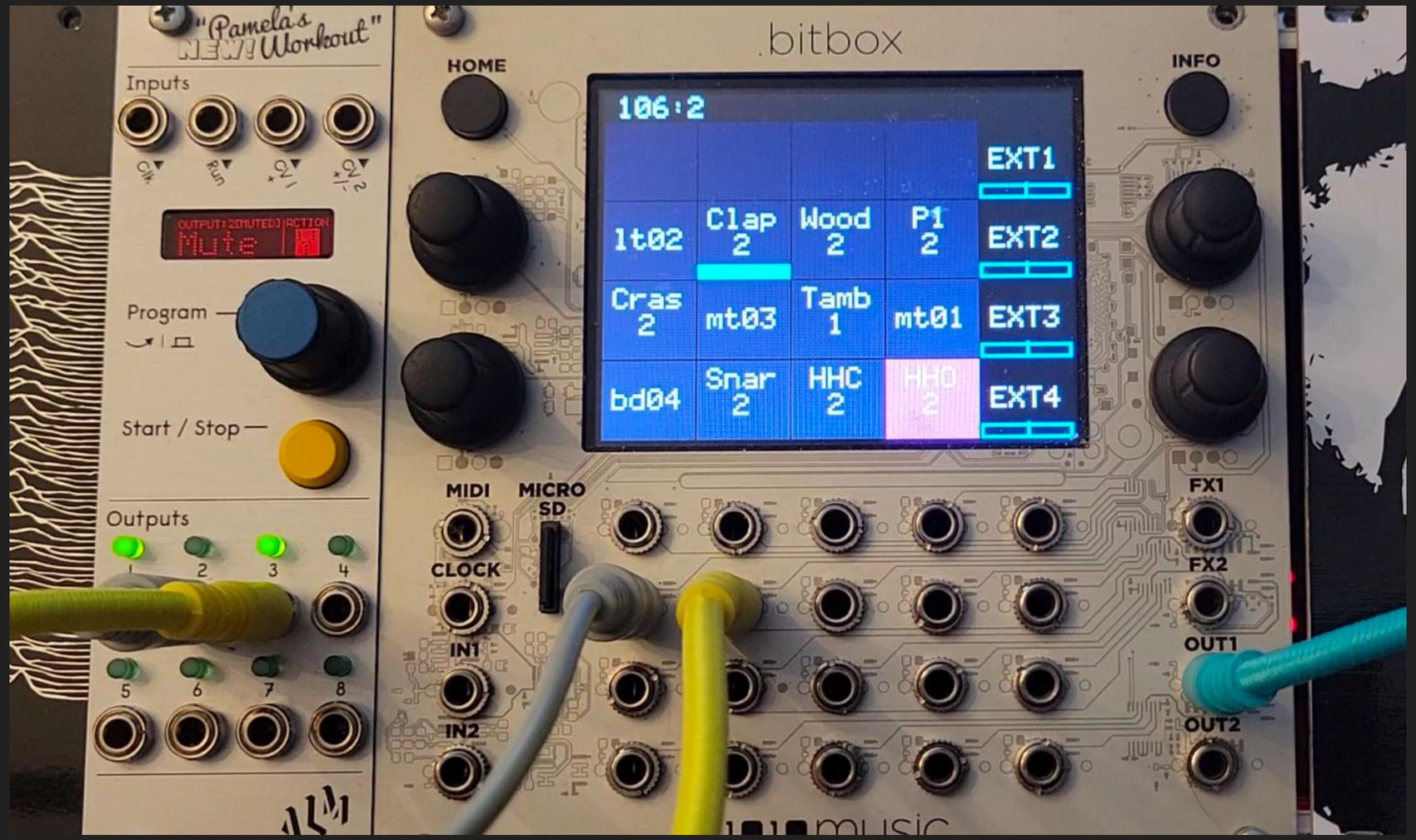
x 2 =



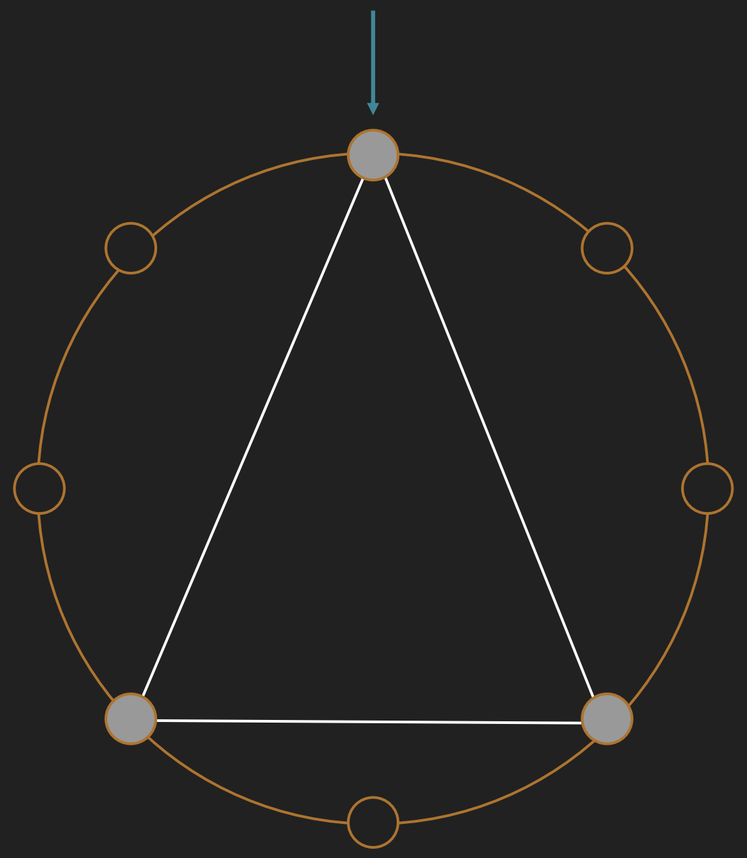
E(4,6,0)

DEMO TIME!

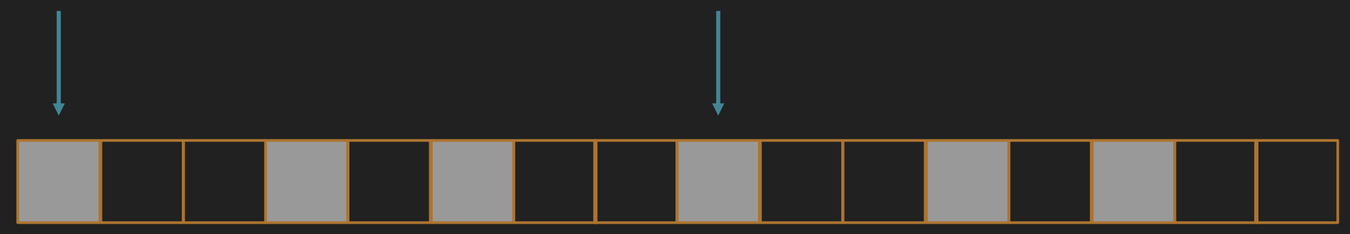
Euclideanistically?



Pams + Bitbox = Full rhythm section



E(3,8,0)



You can overlap the same rhythm but with different rotations

ODD TIME SIGNATURES

Caveat: I don’t really know what I’m talking about.

But when has that stopped someone on YouTube?

Don’t get me wrong, I love modular noodlings, but there’s reason we call it noodling.

Go figure

Though, I suppose there’s a reason 4/4 is called ‘common time’

2/4

2/4

4/4

2/4

4/4

3/4

2/4

4/4

3/4

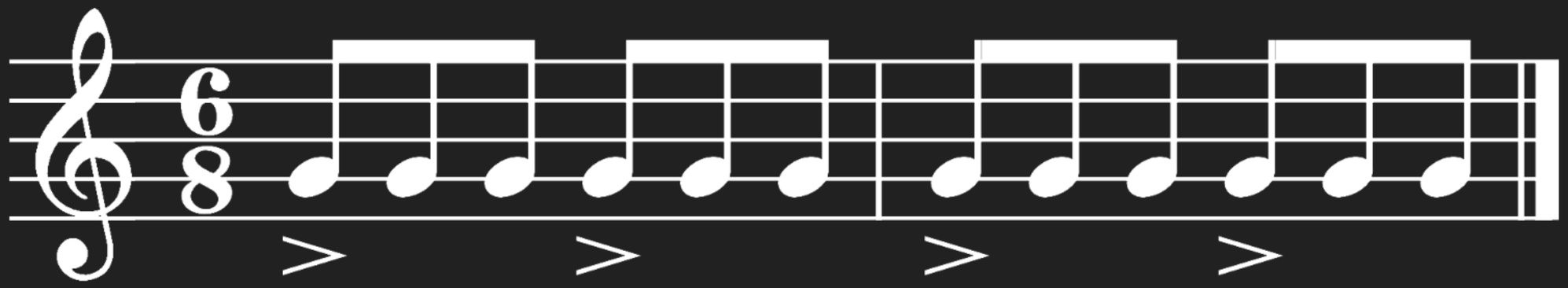
6/8

Such numbers

Wow

Simple meters subdivide beats into groups of two

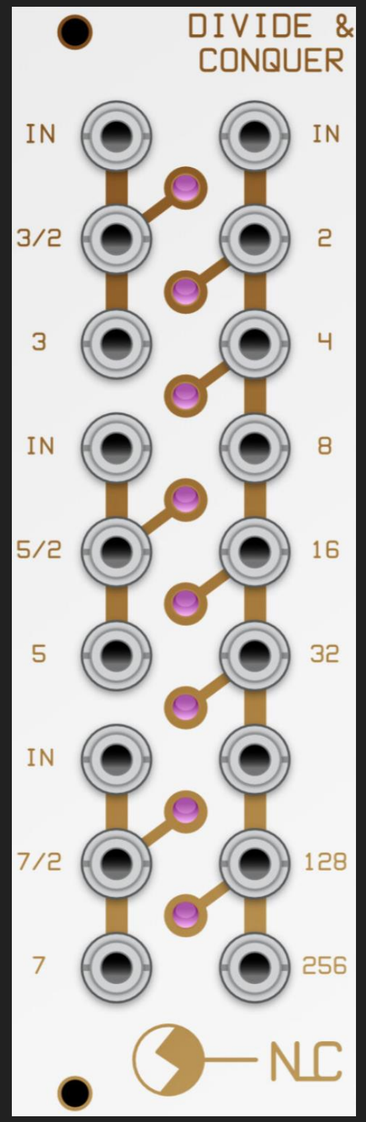
Compound meters break them into groups of three



*This would be ¾ time

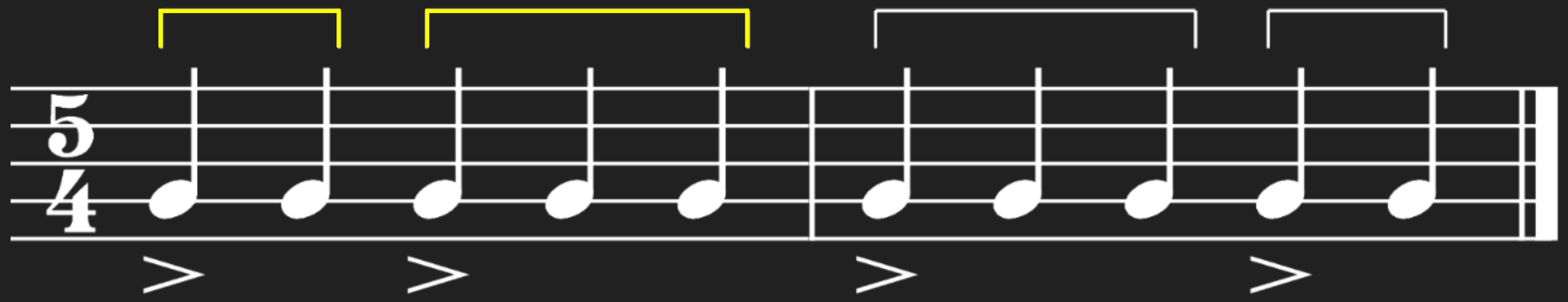
I’m using Pamela’s New Workout for all the Euclidean stuff, and I think I’m going to do a couple of videos about it.

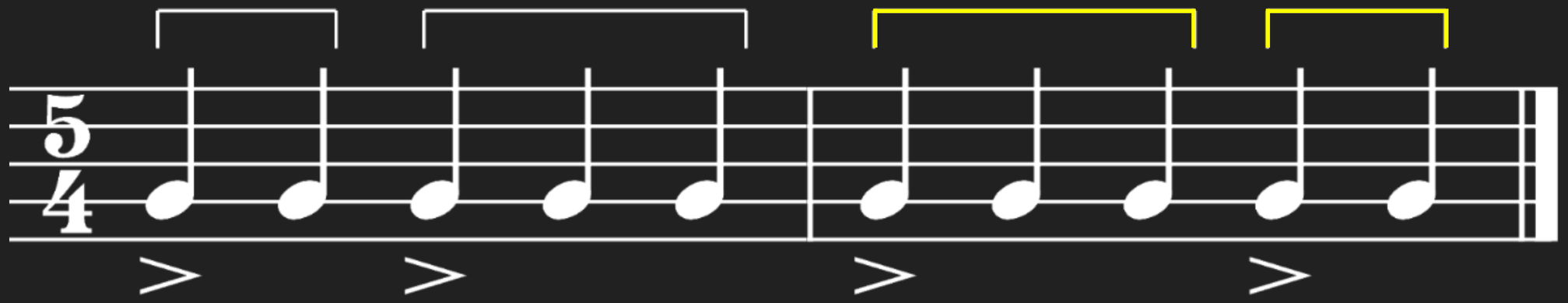
But I honestly prefer dedicated clock dividers.



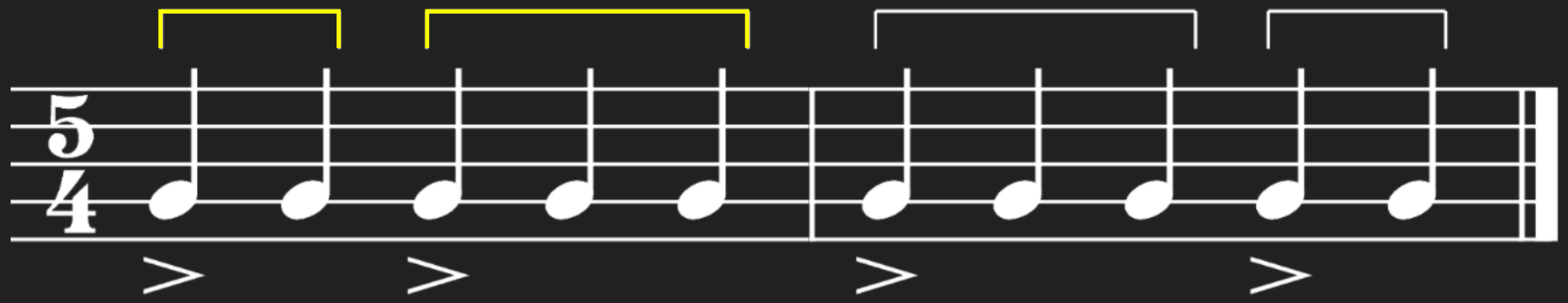
Odd/irregular time signatures cannot be broken up uniformly into groups of 2 or 3

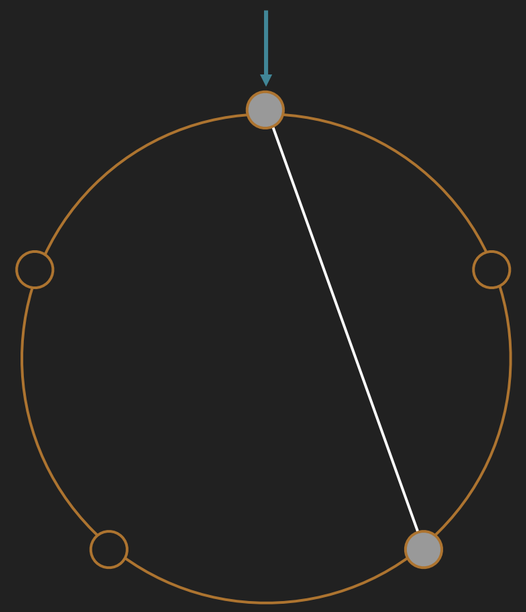
It’s still groups of 2 & 3, but they get mixed in each measure


5
4

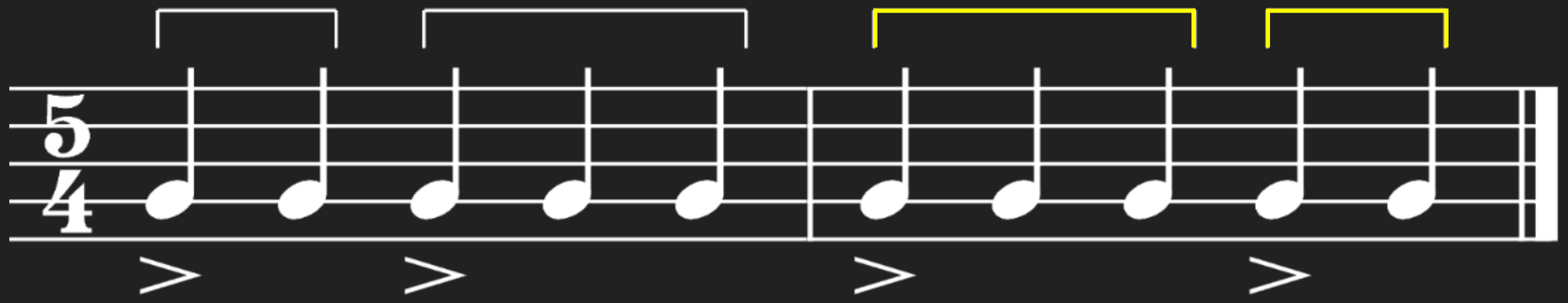

5
4

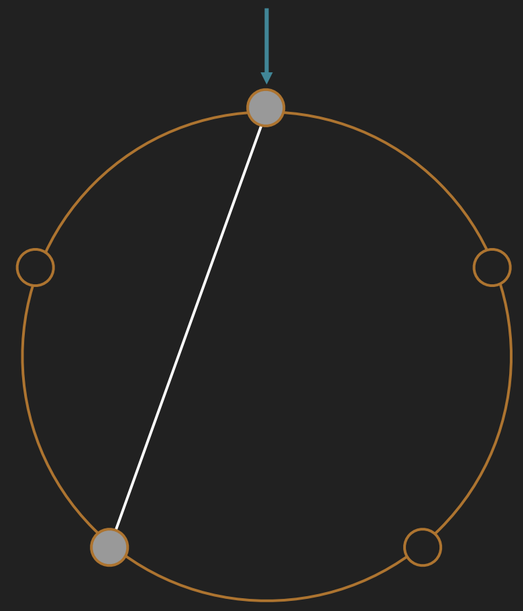
Anytime you want to break time up into uneven groups, Euclid has you covered.



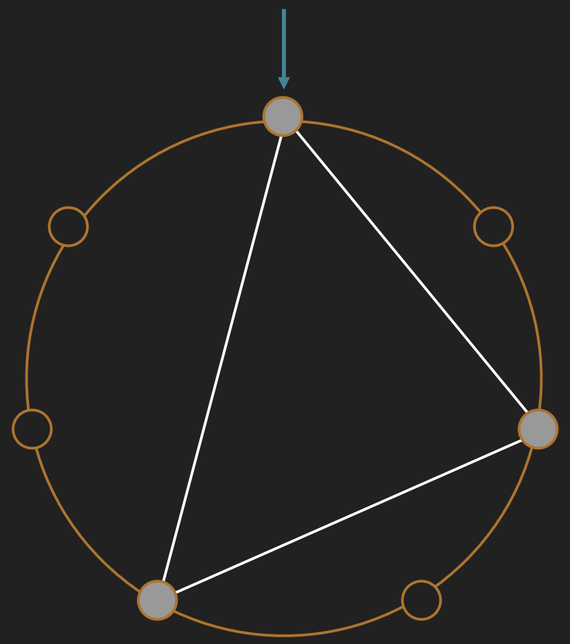


E(2, 5, 0)

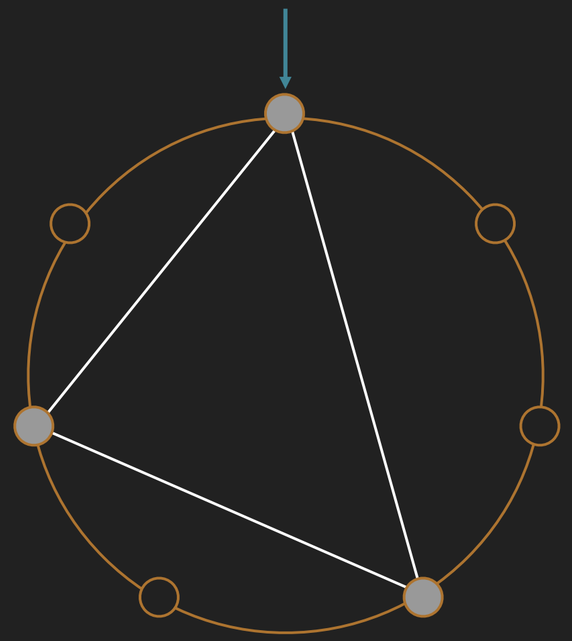




E(2, 5, 2)



E(3, 7, 5)



E(3, 7, 2)

Always looking for different ways to use a module or technique.

Modular is an exercise in problem solving

RHYTHM ZOO

Toussaint found these patterns in music from around the world.

What will you make with them?

Slides at: https://bit.ly/euclideanrhythms

5-step patterns

E(2,5,0)

13th century Persian rhythm *Khafif-e-ramal,* Tchaikovsky's *Symphony No. 6* (2nd Movement)

E(2,5,2)

Dave Brubeck *Take Five,* Gustav Holst, *The Planets (Mars)*

E(3,5,2)

Rumanian folk-dance rhythm

6-step patterns

E(5,6,2)

A popular Arab rhythm pattern *York-Samai*

7-step patterns

E(3,7,0)

Bulgarian folk dance rhythm *Ruchenitza,* Pink Floyd *Money*

E(4,7,0)

Another *Ruchenitza* rhythm

E(5,7,0)

An Arab rhythm *Nawakhat*

8-step patterns

E(3,8,2)

Cuban *Tresillo* pattern

E(5,8,0)

Cuban *Cinquillo* pattern

E(5,8,2)

Spanish *Tango,* 13th century Persian rhythm *Al-saghilal-sani*

E(7,8,0)

A rhythm of the Tuareg people of Libya

9-step patterns

E(4,9,0)

The *Aksak* rhythm of Turkey, Dave Brubeck *Rondo a la Turk*

E(5,9,0)

A popular Arab rhythm called *Agsag-Samai*

E(5,9,2)

A drum pattern used by the Venda in South Africa, a Rumanian folk-dance rhythm

11-step patterns

E(4,11,0)

Frank Zappa *Outside Now*

E(5,11,0)

Mussorgsky *Pictures at an Exhibition*

12-step patterns

E(5,12,0)

Venda clapping pattern of a South African children's song

E(7,12,0)

A common West African bell pattern

16-step patterns

E(5,16,6)

Brazilian *Bossa Nova*

E(7,16,9)

A clapping pattern from Ghana

E(7,16,14)

Brazilian *Samba*

E(9,16,5)

A cowbell pattern in the Brazilian samba

E(9,16,12)

The bell pattern of the Ngbaka-Maibo rhythms of the Central African Republic

24-step patterns

E(11,24,14)

A rhythm necklace of the Aka Pygmies of Central Africa

E(13,24,5)

Another rhythm necklace of the Aka Pygmies of the upper Sangha

Thanks for watching!

(Did you learn something new? Why not subscribe?)